Evolutionary Multi-Objective Neural Architecture Search for Generalized Cognitive Diagnosis Models

1st Shangshang Yang
School of Artificial Intelligence,
Key Laboratory of Intelligent Computing and Signal Processing of Ministry of Education,
Anhui University Hefei, China
yangshang0308@gmail.com

2nd Cheng Zhen, 3rd Ye Tian, 4th Haiping Ma*
Institutes of Physical Science and Information Technology,
Key Laboratory of Intelligent Computing and Signal Processing of Ministry of Education,
Anhui University Hefei, China
zhencheng0208@gmail.com,field910921@gmail.com,hpma@ahu.edu.cn

5th Yuanchao Liu
State Key Laboratory of Synthetical Automation for Process Industries,
Northeastern University Shenyang, China
Yuanchaoliu@126.com

6th Panpan Zhang, 7th Xingyi Zhang*
School of Artificial Intelligence,
Key Laboratory of Intelligent Computing and Signal Processing of Ministry of Education,
Anhui University Hefei, China
zppan55@gmail.com,xyzhanghust@gmail.com

2023 5th International Conference on Data-driven Optimization of Complex Systems (DOCS) | 979-8-3503-9352-1/23/$31.00 ©2023 IEEE | DOI: 10.1109/DOCS60977.2023.10294588

***Abstract*—Cognitive diagnosis models (CDMs) with high generalization are essential for intelligent education systems to reveal students' knowledge states in multiple datasets. However, existing CDMs' architectures are designed dependent on researcher expertise and experience from observing and summarizing partial students' learning behaviors, which makes handcrafted models not cover all learning behaviors well and thus limits their generalization. To develop generalized CDMs, this paper proposes an evolutionary neural architecture search to design CDMs' architectures effective on multiple datasets automatically. Specifically, we first formulate the search task as a multi-objective optimization problem (MOP), which maximizes model performance on multiple datasets containing learning behaviors as many as possible to ensure model generalization. Then, an expressive search space is devised to contain as many potential architectures as possible, where each architecture is denoted by a unified form, taking three given inputs and integrating them in a linear or no-linear manner for prediction. Finally, an evolutionary algorithm with a tailored deduplication strategy is employed to solve the MOP, where each architecture is further represented by a single-root computation tree for easy optimization. Experiments on multiple datasets demonstrate the generalization and effectiveness of the best architecture searched by the proposed approach, and the searched architecture also holds as good interpretability as handcrafted architectures.**

***Index Terms*—Cognitive diagnosis, neural architecture search, evolutionary algorithm, multi-objective optimization**

I. Introduction

Cognitive diagnosis (CD) plays a vital role in intelligent education [1] to estimate students' proficiency in specific knowledge concepts based on students' historical exercising records and the exercise-concept relational matrix (dubbed Q-matrix) [2]. Fig. 1 gives a toy example of CD, where students $\{Alex, Jack\}$ have practiced a set of exercises (i.e., $\{e_1, e_3, e_4\}$ and $\{e_1, e_2, e_3\}$) and got the responses, respectively. Based on the response records and Q-matrix, the CD can diagnose the students' knowledge mastery (shown on the top right). As a result, the students' diagnosis results benefit

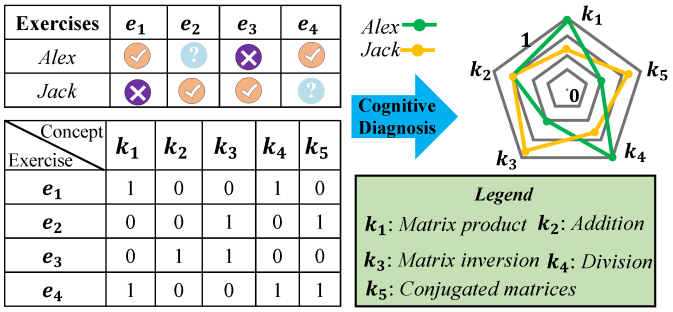


Fig. 1: The illustration of the cognitive diagnosis process. The response records of students $\{Alex, Jack\}$, and the exercise-concept relational matrix (Q-matrix) are shown on the left, while the top right part presents their corresponding diagnosis results.

a wide range of intelligent education tasks, such as targeted training [3] and exercise recommendation [4], [5].

A lot of researchers have developed various CD approaches, which generally lie in two types [6]. The first genre of approaches (e.g., ECD [7], CDGK [8], and PAKP [9]) incorporates more rich input information for enhancing models' input representations, yet directly employs existing diagnostic functions for prediction. The second genre focuses on devising diagnostic functions to integrate given inputs in a highly interpretable manner for prediction, which further contains classical educational psychology-based models (e.g., IRT [10] and DINA [11]) and neural network-based models, such as NeuralCDM [12], RCD [13], and KSCD [6], where KSCD and RCD are equipped with the enhanced input information.

With the rapid increment of student learning data, many datasets with different features are built by different education platforms. However, existing CDMs' design is over-dependence on human domain knowledge [12] and their experience from observing and summarizing partial students'

979-8-3503-9352-1/23/$31.00 ©2023 IEEE

Authorized licensed use limited to: Anhui University. Downloaded on October 26,2026 at 14:40:42 UTC from IEEE Xplore. Restrictions apply.

learning behaviors, where the design bias in researchers' experiences may not only limit the generalization of CDMs when applied to datasets containing unobserved learning behaviors but also limit the emergence of novel CDMs beyond human capabilities. Besides, manually designing CDMs is labor-intensive and needs a lot of trial-and-error, and their architectures are too simple to model complex student learning behaviors [6], [14] in education platforms.

To develop generalized CDMs, this paper proposes to design CDMs' architectures automatically by leveraging neural architecture search (NAS), which aims to find the architectures that hold promising performance on multiple datasets maintaining different learning behaviors and features. As well known, although NAS [15] has made significant progress in many fields to search for different types of neural networks (NNs) [16]–[19], existing NAS approaches cannot be directly applied to CD due to the big difference in search space for NNs between CD and other domains. Furthermore, previous NAS paradigms find good architecture(s) for a single dataset and directly transfer the architectures to other datasets, which is not always effective when being applied to CD due to the significant difference among the datasets collected by different education platforms.

Therefore, we propose designing generalized CDMs using the evolutionary multi-objective optimization-based NAS technique. To be specific, our main contributions are as follows:

- To the best of our knowledge, our work is the first to design generalized CDMs via NAS, termed NAS-GCD. To ensure the high model generalization, we first train the architecture on multiple datasets containing different learning behaviors respectively, then formulate the search task as a multi-objective problem (MOP) to maximize the architecture performance on multiple datasets. Besides, an expressive search space is devised for NAS-GCD, where all architectures can be denoted by a unified form taking three given inputs for prediction.
- To solve the formulated MOP, an effective multi-objective evolutionary algorithm (MOEA) is proposed to explore the search space. Considering the powerful ability of MOEAs in solving tree-encoding problems, each architecture is transformed into a tree-architecture for easy optimization. Besides, an effective deduplication strategy is proposed to speed up the search convergence.
- Extensive experiments on multiple education datasets demonstrate the generalization and effectiveness of the architectures found by NAS-GCD, where the interpretability of the found architecture is also validated.

II. Related Work

In this section, we briefly review the related work on cognitive diagnosis and neural architecture search.

A. Cognitive Diagnosis

The CD task aims to reveal a student's proficiency in specific knowledge concepts based on his response records, where the output prediction is got by a diagnostic function with given inputs, including the student-related vector, the exercise-related vector, and the knowledge concept-related vector. Therefore, a lot of researchers have developed various diagnostic functions from two perspectives.

The first type of diagnostic function is developed from the sight of educational psychology. The classical IRT [10] employs the logistic function combining the linear transformation of student and exercise inputs as its diagnostic function to get the prediction. The diagnostic function of DINA [11] utilizes the inner product to integrate different latent features of student and exercise inputs. MIRT [20] holds the same diagnostic function as IRT yet deals with multidimensional features. MF [21] is based on matrix factorization yet its diagnostic function can be simplified as the inner-product function. The second type of diagnostic function is based on NNs and was invented by researchers in machine learning. The most representative approach is NeuralCDM [12], whose diagnostic function employs one shallow layer inspired by MIRT to linearly combine input features and uses three fully connected (FC) layers to get the output. KSCD's diagnostic function [6] consists of three FC layers followed by one shallow layer same as NeuralCDM, where FC layers are used to boost student and exercise features by combining the concept feature. RCD [13] directly concatenates three inputs enhanced by its GNN in pairs and then gets the final prediction via three FC layers.

As can be seen, manually designing CDMs based on human domain knowledge is labor-intensive and needs a lot of trial-and-error, where the researcher design experience summarized from partial students' learning behaviors cannot cover all learning behaviors and thus limits the model generalization to new datasets. Besides, the human design bias also limits the emergence of novel diagnostic functions beyond human recognition. Therefore, this paper aims to design generalized CDMs by applying NAS to multiple datasets containing various learning behaviors.

B. Neural Architecture Search

As an automated machine learning technique [15], [22], NAS has been widely used for many research domains [23] and made significant progress since first proposed in [24]. Existing NAS approaches have been used for various domains to search the best architectures of various prevailing DNNs, including CNNs for computer vision (CV) tasks [25], RNNs for natural language processing (NLP) [17] and speech-related [18] tasks, GNNs for non-Euclidean data tasks [26], and Transformers for CV [27], NLP [28], and speech-related [29] tasks.

However, existing NAS approaches cannot be directly applied to CD for designing effective diagnostic function architectures since their search space is not applicable to CD. Besides, current NAS approaches in CV and NLP commonly search for the best architecture for a single dataset, where the architecture generally holds competitive performance on other datasets [23], but it may not always be effective when applying to CD due to the significant difference among different education datasets. Inspired by the pioneer NAS work [30] for

Authorized licensed use limited to: Anhui University. Downloaded on October 26,2026 at 14:40:42 UTC from IEEE Xplore. Restrictions apply.

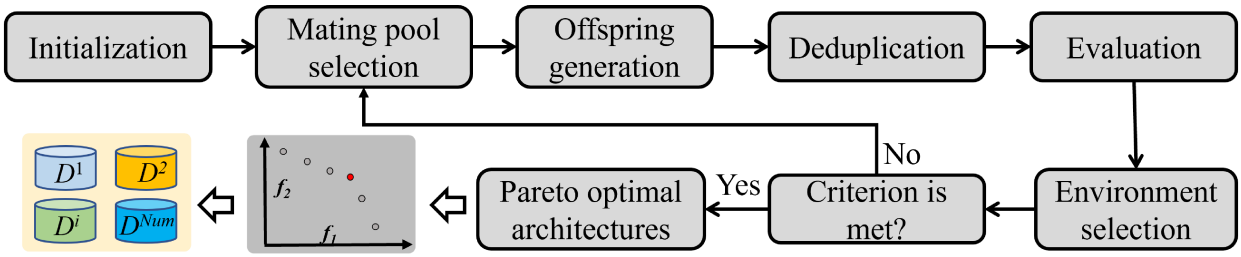


Fig. 2: The overview of the proposed NAS-GCD, which searches and selects the best architecture for CD on multiple datasets.

CD, our work aims to leverage the NAS technique to design generalized CDMs for multiple datasets.

III. Preliminaries

In this section, we first present the problem definition of the CD task, and then summarize the general formulation of CD.

A. Cognitive Diagnosis Task

To implement the cognitive diagnosis task, there commonly store N students, M exercises, and K knowledge concepts in a modern intelligent learning system platform, which can be denoted as $S = \{s_1, s_2, \cdots, s_N\}$, $E = \{e_1, e_2, \cdots, e_M\}$, and $C = \{c_1, c_2, \cdots, c_K\}$, respectively. Generally, each student has practiced a set of exercises and got corresponding scores, and thus there form the students' response logs $R_{log} = \{(s_i, e_j, r_{ij}) | s_j \in S, e_j \in E, \text{and } r_{ij} \in \{0, 1\}\}$, where $r_{ij} = 1$ indicates the answer of student s_i on e_j is correct and $r_{ij} = 0$ otherwise. In addition, there is a Q-matrix $Q = (Q_{jk} \in \{0, 1\})^{M \times K}$, where $Q_{jk} = 1$ if exercise e_j contains knowledge concept c_k and $Q_{jk} = 0$ otherwise.

With the above students' logs R_{log} and Q-matrix, the goal of the cognitive diagnosis task is to reveal students' proficiency in specific knowledge concepts by predicting students' response scores based on a model $\mathcal{F}$.

B. General Formulation of CD

In general, the model $\mathcal{F}$ (i.e., the diagnostic function) receives three given types of inputs and outputs the predicted response score $\hat{r}_{ij}$ of student S_i got on exercise E_j as

$$\hat{r}_{ij} = \mathcal{F}(\mathbf{h}_S, \mathbf{h}_E, \mathbf{h}_C), \tag{1}$$

where $\mathbf{h}_S \in R^{1 \times D}$ is the student-related feature vector, $\mathbf{h}_E \in R^{1 \times D}$ is the exercise-related feature vector, $\mathbf{h}_C \in R^{1 \times K}$ is the knowledge concept-related feature vector and D is the embedding dimension (usually equal to K). The three vectors can be obtained via the following process:

$$\begin{aligned} \mathbf{h}_S &= \mathbf{x}_i^S \times W_S, W_S \in R^{N \times D} \\ \mathbf{h}_E &= \mathbf{x}_j^E \times W_E, W_E \in R^{M \times D} \\ \mathbf{h}_C &= \mathbf{x}_j^E \times Q = (Q_{j1}, Q_{j2}, \cdots, Q_{jK}) \end{aligned}, \tag{2}$$

where $\mathbf{x}_i^S \in \{0, 1\}^{1 \times N}$ and $\mathbf{x}_j^E \in \{0, 1\}^{1 \times M}$ are one-hot vectors for student s_i and exercise e_j, respectively, and W_S and W_E are trainable matrices in the embedding layers.

Algorithm 1 Main Steps of NAS-GCD.

Input: $Maxi_Eva$: Maximum number of evaluations; Pop: Population size;
Output: P_{non}: Non-dominated individuals;
$P \leftarrow Initialization(Pop)$, $S_{table} \leftarrow \emptyset$;
Obtain the objective values of each individual in P by training its encoded architecture according to (5);
while $Maxi_Eva$ *is not met* **do**
  P'←Select Pop parent individuals from population P;
  Q←Genetic Operation(P');
  Q, S_{table} ←Deduplication(Q, P', S_{table}, Pop);
  Obtain the objective values of each individual in Q by training its encoded architecture according to (5);
  P←Environment selection($P \cup Q$);
end
Select non-dominated individuals P_{non} from P as the output;
Result: P_{non}

IV. The Proposed NAS-GCD

This section will first present the proposed NAS-GCD framework, and then sequentially give the suggested search space for CD, the formulated MOP for the NAS task of CD on multiple datasets, the encoding strategy, the genetic operator for offspring generation, and the solution deduplication strategy for deleting duplicates and saving computation costs.

A. Overview

For better comprehension, the overall framework of NAS-GCD is presented in Fig. 2, which is basically the same as the common EA consisting of six main steps.

To start with, a population P comprising Pop individuals is randomly initialized from the search space. To facilitate rapid convergence and prevent starting the search process entirely from scratch, some individuals' architectures are initialized from handcrafted architectures [31], [32]. Second, to select good potential individuals for offspring generation, the standard binary tournament selection [33] is applied to the population P, obtaining the mated individuals to form the mated population P'. Next, to generate effective offspring individuals and obtain the offspring population Q, a tailored genetic operator (as described in Section IV-E) is applied to the mated population P', where one of four operations is randomly used. Fourthly, an effective solution deduplication strategy (in Section IV-F) is devised to delete duplicated individuals to

Authorized licensed use limited to: Anhui University. Downloaded on October 26,2026 at 14:40:42 UTC from IEEE Xplore. Restrictions apply.

TABLE I: 15 Candidate Operators in The Proposed Search Space.

Notation	Syntax	Output shape
Unary operators		
Neg	$-x$	same as x
Abs	$\|x\|$	same as x
Inv	$1/(x+\epsilon), \epsilon = 1e^{-6}$	same as x
Square	x^2	same as x
Sqrt	$\text{sign}(x) \cdot \sqrt{\|x\| + \epsilon}$	same as x
Tanh	$tanh(x)$	same as x
Sigmoid	$sigmoid(x)$	same as x
Softplus	$softplus(x)$	same as x
Sum	$\sum_i^D \mathbf{x}_i$	R^1
Mean	$(\sum_i^D \mathbf{x}_i)/D$	R^1
FFN	$\mathbf{x}_i \times W_F, W_F \in R^{D\times 1}$	R^1
FFN_D	$\mathbf{x}_i \times W_F, W_F \in R^{D\times D}$	$R^{1\times D}$
Binary operators		
Add	$x + y$	$max(x, y)$
Mul	$x \cdot y$ or $x \odot y$	$max(x, y)$
Concat	$[\mathbf{x}, \mathbf{y}] \times W_{Con}, W_{Con} \in R^{2*D\times D}$	$R^{1\times D}$

save computation costs (denoted by Deduplication). Fifthly, evaluate the objective values in (5) of each individual in Q by training the architecture encoded by each individual for a certain number of epochs. Afterward, the environmental selection in NSGA-II [34] will be employed to identify and maintain the individuals that hold better objective values from the union of population P and offspring population Q.

The second to the sixth step will be repeated until the maximal number of evaluations is exceeded. Finally, the non-dominated individuals P_{non} in the final population P will be output as the set of Pareto optimal individuals. For more details, Algorithm 1 summarizes the main procedures of the proposed NAS-GCD. After the end of the proposed NAS-GCD, one individual in the knee area [35], [36] will be selected from P_{non} for the subsequent CD task.

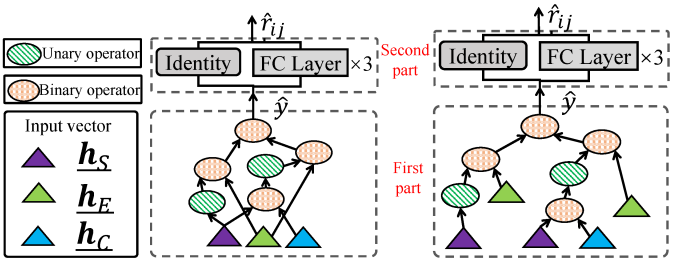
B. Search Space for CD

Based on the general formulation of CD in (1), the diagnostic function architecture is used to combine three given types of feature vectors ($\mathbf{h}_S$, $\mathbf{h}_E$, and $\mathbf{h}_C$) in a linear or nonlinear manner and outputting the prediction. Therefore, we propose a novel and expressive search space, where each candidate diagnostic function architecture A consists of two parts.

The first part $\mathcal{M}_1$ takes $\mathbf{h}_S$, $\mathbf{h}_E$, and $\mathbf{h}_C$ as its input and outputs a scalar or a vector, while the second part $\mathcal{M}_2$ is an identity operation or three FC layers. The overall forward processes of two parts are as follows:

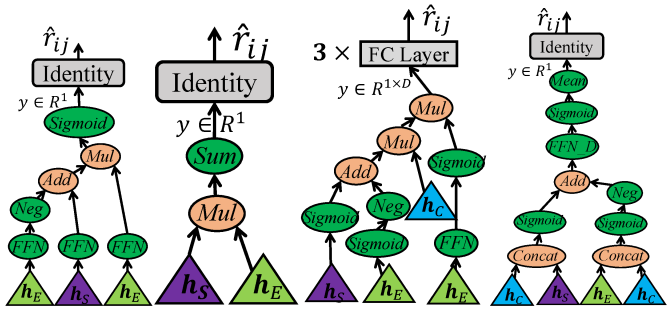
$$\begin{aligned} &\hat{r}_{ij} = \mathcal{M}_2(\hat{y}),\ \hat{y} = \mathcal{M}_1(\mathbf{h}_S, \mathbf{h}_E, \mathbf{h}_C) \\ &\mathcal{M}_2(\cdot) = \begin{cases} Identity(\cdot), \text{if } \hat{y} \in R^1 \\ FC_3(FC_2(FC_1(\cdot))), \text{if } \hat{y} \in R^{1\times D} \end{cases}. \end{aligned} \tag{3}$$

To contain a large quantity of good potential architectures, $\mathcal{M}_1$ is set to consist of input nodes and some computation nodes, where the number of computation nodes, the links of nodes, and the computation nodes' operators determine the architecture of $\mathcal{M}_1$ and the architecture of A. Inspired by existing CDMs [12], Table I collects 15 primitive operators in two types, i.e., unary operators and binary operators, as the candidate operators for computation nodes.



(a) Original architecture. (b) Tree-like architecture.

Fig. 3: The illustration of a diagnostic function architecture under the proposed search space in (a), which can be represented by the tree-like architecture in (b) for easy optimization in MOEAs.



(a) IRT. (b) MF. (c) NeuralCDM. (d) RCD.

Fig. 4: Four typical CDMs plotted under the devised search space.

For better understanding, Fig. 3(a) plots a general diagnostic function architecture under the proposed search space. It is obvious that the whole architecture A is mainly determined by its first part, which is further affected by its topology and contained computation nodes.

According to previous experiences [23], an effective search space should contain not only a large number of potential architectures but also existing handcraft architectures as many as possible. To demonstrate this, we present four representative CDMs including IRT, MF, NeuralCDM, and RCD under the proposed search space, whose architectures are shown in Fig. 4. Therefore, the effectiveness of the proposed search space can be proven to some extent.

C. Formulated MOP

To make the search architecture effective on multiple (Num) datasets $\mathcal{D} = \{D^i = \{D^i_{train}, D^i_{val}, D^i_{test}\} | 1 \le i \le Num\}$, the architecture search for CD can be formulated as

$$\max_{A\in\mathcal{A}} \mathbf{F}(A) = \begin{cases} f_{val_AUC}(A, D^1_{val}) \\ f_{val_AUC}(A, D^2_{val}) \\ \cdots \\ f_{val_AUC}(A, D^{Num}_{val}) \end{cases}, \tag{4}$$

where $f_{val_AUC}(A, D^i_{val})$ denotes the validation AUC (*Area Under an ROC Curve*) [37] value of architecture A on i-

Authorized licensed use limited to: Anhui University. Downloaded on October 26,2026 at 14:40:42 UTC from IEEE Xplore. Restrictions apply.

Algorithm 2 Deduplication Strategy.

Input: Q: Offspring population; P': A mated population; S_{table}: A table that stores the ordered stacks of all previous individuals; Pop: Population size;
Output: P_{ded}:Population individuals after deduplication;S_{table}

$P_{ded} \leftarrow \emptyset$;
while $|P_{ded}| < Pop$ **do**
  for *i=1 to* $|Q|$ **do**
    $Q_i \leftarrow Q[i]$; // *i-th individual in* Q
    $Q_i^m \leftarrow$Standardize(Q_i); // *The standardized tree*
    $Q_i^s \leftarrow$Serialize(Q_i^m); // *The serialized representation*
    if Q_i^s *not in* S_{table} **then**
      $P_{ded} \leftarrow Q_i^m$, $S_{table} \leftarrow Q_i^s$;
    end
    if $|P_{ded}| \geq Pop$ **then**
      break;
    end
  end
  $Q \leftarrow$Genetic Operation(P');
end
return P_{ded}, S_{table}

th dataset after training A on D_{train}^i. Note that the above-formulated problem is a MOP and there exist conflicts between some objectives due to the difference between different datasets in terms of distribution, attribute, and so on.

For easy final solution selection, the above MOP can be simplified as

$$\max_{A \in \mathcal{A}} \mathbf{F}(A) = \begin{cases} \frac{\sum_i^{Num} f_{val_perf}(A, D_{val}^i)}{Num} \\ 1 - Std(\{f_{val_perf}(A, D_{val}^i) | 1 \leq i \leq Num\}) \end{cases}, \tag{5}$$

where the first item maximizes the performance of architecture A averaged on Num datasets and the second item minimizes the standard deviation of its performance on multiple datasets to avoid the bias of performance on easy or difficult datasets. By doing so, we can search the optimal architecture on multiple datasets for CD by solving the above problem.

However, there are two challenges in solving the above problem for searching good diagnostic function architectures on multiple datasets. First, the problem in (5) is a MOP, which cannot be solved well by common NAS search strategies, such as gradient-based optimization techniques. Second, candidate architectures in the search space hold variable-number nodes, which poses a challenge in optimizing the node number, the architecture topology, and the computation nodes' operators.

D. Encoding Strategy

The above-mentioned first issue can be easily addressed by proposing and applying an effective multi-objective evolutionary algorithm to the MOP. To overcome the second obstacle, considering the particularity of the first part $\mathcal{M}_1$ having only one output, the general diagnostic function architecture can be further represented by a single-root computation tree in Fig. 3(b). By doing so, the original search task with variable-number nodes can be transformed into the tree-architecture search task, which is easier to solve by MOEAs equipped with specific genetic operators.

As a result, the proposed NAS-GCD directly adopts the tree-based encoding (as shown in Fig. 3(b)) for each individual to encode the neural architecture in the proposed search space. Compared to vector-based encoding, tree-based encoding is easier for optimization since some optimization difficulties (e.g., the variable-length encoding problem) can be avoided.

E. Genetic Operator

As introduced above, all architectures in the search space are transformed into tree-architectures and then encoded by trees in the MOEA. Due to the nature of tree-encoding, typical genetic operators (e.g., single-point crossover and simulated binary crossover [38]) are inapplicable. For this aim, we devise a genetic operator based on four modified operations *Exchange*, *Delete*, *Insert*, and *Delete*, which are popular in genetic programming [39]. Fig. 5 gives illustrative examples of four operations and their descriptions are as follows:

- *Exchange*. Exchange two random subtrees of two individuals to form new offspring individuals, where their root nodes will not be selected.
- *Delete*. Randomly delete a node in an individual by connecting one of the left and right trees to its parent node.
- *Insert*. Insert a new node holding a new sampled operator between a randomly selected node and its parent node. Adding a new leaf node sampled from $\{\mathbf{h}_S, \mathbf{h}_E, \mathbf{h}_C\}$ if the new node's operator is binary.
- *Replace*. Replace the operator of a randomly selected node in an individual with a newly sampled operator. Randomly add a new leaf node to the selected node if the new sampled operator needs more inputs and randomly keep one of its left and right child trees otherwise.

With the above four operations, the devised genetic operator will randomly select one of four operations each time and apply it to two individuals to generate two offspring individuals until all offspring individuals are generated.

F. Solution Deduplication Strategy

Due to the simplicity of tree-architectures, there may exist a large number of duplicate architectures during the evolution. To improve search efficiency and accelerate search convergence of the proposed NAS-GCD, we propose a solution deduplication strategy to identify and delete duplicate architectures. Considering the symmetry of some operators including *Add*, *Mul*, and *Concat*, it is vital for each individual's architecture to first rearrange the left and right subtrees of the nodes containing the above operators, then obtain the rearranged architecture for each architecture, and finally remove the architectures with the same arranged architecture.

To begin with, the notations in Table I and $\{Stu, Exce, Conc\}$ are used as the identifiers of the computation nodes and leaf nodes in each individual's

Authorized licensed use limited to: Anhui University. Downloaded on October 26,2026 at 14:40:42 UTC from IEEE Xplore. Restrictions apply.

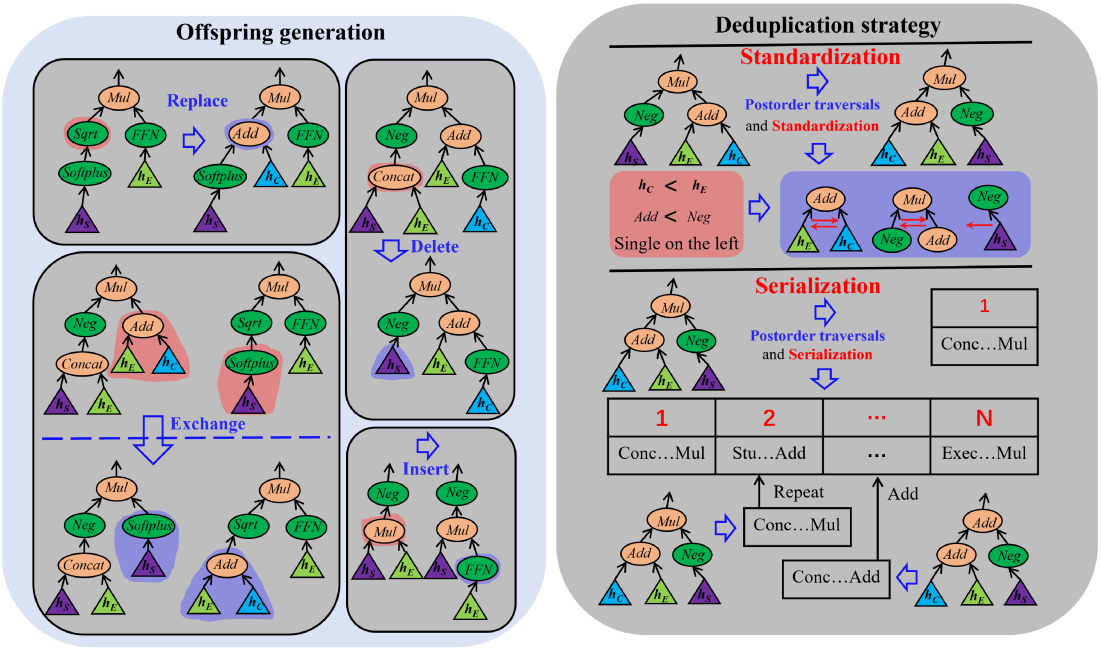


Fig. 5: The proposed Genetic Operator and Deduplication Strategy.

architecture, i.e., each individual can be regarded as an ordered stack of these identifiers. Based on a given table S_{table} that stores the ordered stacks of all previous individuals, we execute the following steps for the architecture encoded by each individual:

1) **Standardization**. Rearrange each individual according to the following rules: during executing the postorder-traverse for the individual, if a node has only one leaf node, put the leaf node as its left node; If a node has two leaf nodes, compare the ASCII[1] values of the two leaf nodes' identifiers, and then take the smaller one and the larger one as its left node and its right node, respectively.
2) **Serialization**. After obtaining the rearranged architecture for the individual, re-executing the postorder-traverse for the individual to obtain the serialized representation (i.e., an ordered stack of identifiers) of this individual.
3) Query the stored sequence table S_{table}: discard this individual if the serialized representation of this individual has appeared in S_{table}, otherwise, add the serialized representation to the table S_{table}.
4) If the size of the population P_{ded} after deduplication is less than Pop, more offspring individuals will be generated based on P', and repeat the above three steps until its population size is met.

We demonstrate the specific and detailed process in Algorithm 2. For better understanding, the top left of Fig. 5 gives a toy example of the deduplication strategy.

V. Experiment

The experiments conducted on multiple real-world datasets are used to answer the following research questions:

- **RQ1**: How about the performance of architectures found by NAS-GCD compared to state-of-the-art CDMs?
- **RQ2**: How about the found architectures' generalization on the datasets that are unobserved in the search phase?
- **RQ3**: How about the effectiveness of the proposed deduplication strategy?
- **RQ4**: How about the interpretability of the architectures found by NAS-GCD?

A. Experiment setup

1) Datasets: To verify the effectiveness of NAS-GCD, we use one private dataset e-Math and four public datasets Junyi [40], SLP [41], ASSISTments2009 [42], and ASSISTments2012 [43], whose descriptions are as follows:

- Junyi: A public dataset collected from Junyicontains 200K interactive records of about 1,000 students.
- SLP: Published by AICFE in 2021, containing nearly 58,000 logs from 907 exercises for 1,499 students.
- e-Math: Collected by a well-known e-education product and consisting of the records of primary and secondary school students' math exercises and exams.
- ASSISTments2009: Created by the ASSISTments online tutoring system, containing nearly 4,000 students, nearly 18,000 exercises, and over 300,000 response logs.
- ASSISTments2012: Collected by the ASSISTments during 2012 2012-2013 academic year and containing about 2 million interactive records.

In NAS-GCD, three smaller datasets Junyi, SLP, and e-Math are used to search good architectures (Num=3), and other two larger datasets are used in **RQ2** to validate the generalization of searched architectures. According to previous work [12],

[1]https://en.wikipedia.org/wiki/ASCII

Authorized licensed use limited to: Anhui University. Downloaded on October 26,2026 at 14:40:42 UTC from IEEE Xplore. Restrictions apply.

[13], the students with less than 15 response logs are filtered out for all datasets to ensure enough data for modeling.

2) Baseline and Metrics: To show the superiority of architectures found by NAS-GCD, we compare them with five representative CDMs, including DINA [11], IRT [10], and MIRT [20], NeuralCDM [12], and RCD [13], where RCD uses extra enhanced input features and thus the results on some datasets cannot be obtained (denoted by '-' in the following tables). Their descriptions can be found in Section II-A. Besides, we adopt AUC, ACC (*accuracy*), and RMSE (*root mean squared error*) as evaluation metrics.

3) Parameter Settings: The dimension D is equal to K, and the output dimensions of three FC layers are set to 512, 256, and 1.

Search Settings During the search of NAS-GCD, the response logs of each student in each dataset are randomly divided into 70%, 10%, and 20% for training, validating, and testing. The architecture of each individual is trained by Adam [44] with a learning rate of 0.001 to minimize the cross entropy loss [45] between the predictions and targets, where the batch size is 128 and the number of training epochs is 30. For NAS-GCD, the population size Pop is set to 100, and the maximum number of evaluations $Maxi_Eva$ is set to 2,000.

Training Settings The splitting setting of 80% and 20% is used for training and testing, the found architectures are retrained for 50 epochs and others keep the same as the above.

B. Performance Comparison (RQ1)

To demonstrate the superiority of the architectures searched by NAS-GCD, we select the best architecture from P_{non} (denoted as *NAS-GCD-best*) and compare it with comparison CDMs on SLP, Junyi, and e-math, where the averaged ACC, RMSE, and AUC on 10 runs are summarized in Table II.

We can find that the architecture found by NAS-GCD holds the significantly best performance on SLP and e-Math, which holds a performance leading of 0.03 and 0.02 over the state-of-the-art handcrafted model NeuralCDM. Although the performance on Junyi of *NAS-GCD-best* is only slightly better than that of RCD, architecture *NAS-GCD-best* only uses original inputs while RCD utilizes extra input information to enhance its performance, which leads to high computational complexity. When compared to the CDMs that do not use such enhanced input information, the performance difference between *NAS-GCD-best* and these CDMs is more significant.

C. Performance on Unobserved Datasets (RQ2)

Since NAS-GCD aims to design a generalized CDM for multiple datasets, it is necessary to validate the performance of its found architectures on the unobserved datasets.

To this end, Table III presents the performance of the architecture *NAS-GCD-best*, as well as five CDMs on two larger and unobserved datasets ASSISTment2009 and ASSISTment2012. It can be seen that the performance of *NAS-GCD-best* is still better than handcrafted CDMs. Furthermore, it is worth noting that the AUC value of *NAS-GCD-best* is about 0.02 higher than RCD on Assistment2009. Due to the needing for extra information, the generalization of RCD is limited when applied to other datasets.

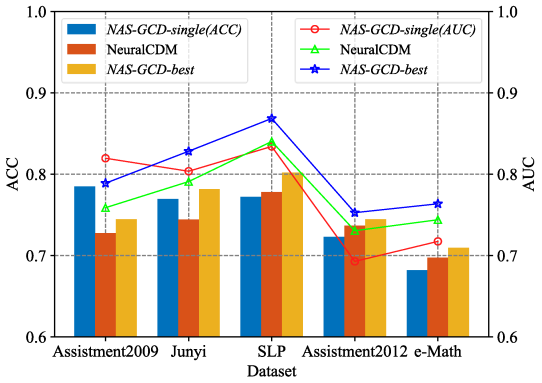


Fig. 6: The performance comparison among *NAS-GCD-best*, NeuralCDM, and *NAS-GCD-single* on five datasets.

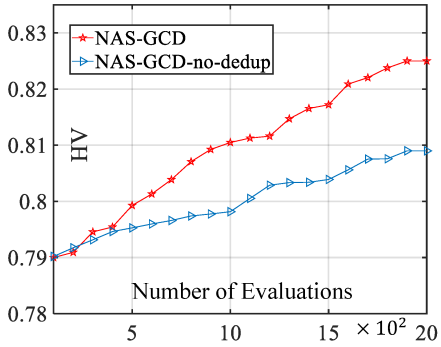


Fig. 7: Effectiveness validation of deduplication strategy by comparing HV convergence curves.

Besides, we build a variant of NAS-GCD to search the optimal architecture on a single dataset (here ASSISTment2009 is used). Fig.6 plots the ACC and AUC values of its found architecture (termed *NAS-GCD-single*), *NAS-GCD-best*, and NeuralCDM on all datasets. We can find that *NAS-GCD-single* holds extremely good performance on its target dataset yet relatively poor performance on other datasets, while *NAS-GCD-best* holds very promising performance on all datasets. As a result, the transferability and generalization of architectures found by NAS-GCD can be demonstrated.

D. Effectiveness of Deduplication Strategy (RQ3)

To demonstrate the effectiveness of the deduplication strategy, Fig. 7 compares the convergence profiles of hypervolume (HV) [46], [47] obtained by NAS-GCD and NAS-GCD-no-dup (NAS-GCD without the strategy), where a larger HV value denotes a better convergence result. We can observe that the employed deduplication strategy can indeed speed up the convergence and lead to better final convergence.

E. Case Study of Architecture NAS-GCD-best (RQ4)

Before answering **RQ4**, Fig. 8 first plots the best architecture found by NAS-GCD. As can be seen, the architecture of *NAS-GCD-best* is relatively complex compared to NeuralCDM

Authorized licensed use limited to: Anhui University. Downloaded on October 26,2026 at 14:40:42 UTC from IEEE Xplore. Restrictions apply.

TABLE II: The performance comparison between the best architecture found by NAS-GCD and five representative CDMs on three datasets in terms of ACC, RMSE, and AUC, averaged on 10 runs. The best results are in bold, and the second-best results are underlined.

Datasets	**SLP**			**Junyi**			**e-Math**		
Metrics	**ACC↑**	**RMSE↓**	**AUC↑**	**ACC↑**	**RMSE↓**	**AUC↑**	**ACC↑**	**RMSE↓**	**AUC↑**
DINA	0.5864	0.5216	0.6561	0.7021	0.4612	0.7364	0.6547	0.4857	0.6853
IRT	0.688	0.4543	0.7711	0.676	0.4268	0.775	0.6723	0.4539	0.6978
MIRT	0.7369	0.4768	0.7852	0.7513	0.4117	0.7989	0.6613	0.477	0.7018
NeuralCDM	0.7779	0.4009	0.8402	0.7443	0.4172	0.7909	0.6975	0.4486	0.744
RCD	-	-	-	0.7716	0.3963	0.8262	-	-	-
NAS-GCD-best	**0.8018**	**0.3759**	**0.8685**	**0.7815**	**0.3879**	**0.8282**	**0.7066**	**0.4338**	**0.7629**

TABLE III: The performance comparison on *NAS-GCD-best* and five CDMs on ASSISTment2009 and ASSISTment2012, where the results are averaged on 10 runs and the best results are in bold.

Datasets	**ASSIST09**			**ASSIST12**		
Metrics	**ACC↑**	**RMSE↓**	**AUC↑**	**ACC↑**	**RMSE↓**	**AUC↑**
DINA	0.6751	0.4694	0.7346	0.7072	0.4465	0.6768
IRT	0.6751	0.4694	0.7346	0.7065	0.4444	0.6776
MIRT	0.7063	0.4515	0.7278	0.7133	0.4431	0.6854
NeuralCDM	0.7277	0.4318	0.7588	0.7367	0.4247	0.7305
RCD	0.7355	0.4213	0.7721	-	-	-
NAS-GCD-best	**0.7447**	**0.4201**	**0.7889**	**0.7446**	**0.4178**	**0.7527**

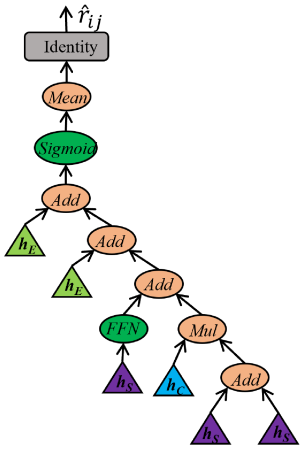


Fig. 8: The best architecture found by the proposed NAS-GCD.

and RCD, which thus provides a more effective manner to integrate input vectors for complex student-exercising data. Here the internal vector of *NAS-GCD-best* involves with $\mathbf{h}_S$ yet not with $\mathbf{h}_E$ denotes the student's knowledge proficiency, while the knowledge difficulty can be denoted by its internal vector involved with $\mathbf{h}_E$ but not with $\mathbf{h}_S$.

On the basis of the above consideration, Fig. 9 depicts an example of a student's diagnosis result obtained by *NAS-GCD-best* on dataset ASSISTments2009, where the upper part gives the Q-matrix and the student's response records, the

	C_{13}	C_{14}	C_{40}	C_{47}	C_{54}	Response
Exercise #2467	1	1	0	0	0	✗
Exercise #7665	0	0	1	1	0	✓
Exercise #10008	0	0	0	0	1	✗

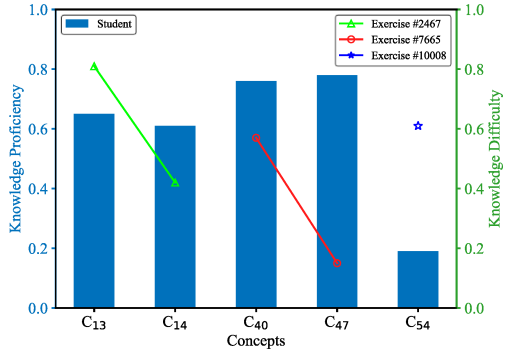


Fig. 9: Case study: Exercise #number refers to exercise code and C_{number} refers to knowledge concept. Here three exercises are randomly selected for illustrative purposes.

bars denote the student's mastery on each concept, and the lines represent the knowledge difficulties of three exercises on relevant knowledge concepts. It can be found that the student answers correctly when his diagnosed knowledge proficiency on one exercise is higher than the obtained knowledge difficulty of the exercise, which satisfies the requirement of CDMs' interpretability. To sum up, we can conclude that the architecture found by the proposed NAS-GCD is reasonable and its diagnosis results for students are also interpretable.

VI. Conclusion

In this paper, we proposed to automatically design generalized CDMs' architectures on multiple datasets that hold different learning behaviors by NAS. Specifically, we first formulated the search task as MOP to maximize model performance on multiple datasets to ensure the model generalization. Then, an expressive search space is devised, where each architecture can be denoted as a single root computation tree. Finally, an effective MOEA with the tree-encoding-specific genetic operator is employed to solve the MOP, where an effective deduplication strategy is proposed to speed up the search convergence. Extensive experiments on multiple

Authorized licensed use limited to: Anhui University. Downloaded on October 26,2026 at 14:40:42 UTC from IEEE Xplore. Restrictions apply.

datasets show the superiority of architectures found by NAS-GCD to handcrafted CDMs, and the interpretability of the found architecture has been also discussed.

ACKNOWLEDGMENTS

This work was supported in part by the National Key Research and Development Program of China (No.2018AAA0100100), and in part by the National Natural Science Foundation of China (No.61876162, No.61906001, No.62136008, No.62276001, No.U21A20512).

REFERENCES

[1] A. Anderson, D. Huttenlocher, J. Kleinberg, and J. Leskovec, "Engaging with massive online courses," in *Proceedings of the 23rd International Conference on World Wide Web*, 2014, pp. 687–698.

[2] K. VanLehn, "Student modeling," *Foundations of intelligent tutoring systems*, vol. 55, p. 78, 1988.

[3] J. Beck, "Difficulties in inferring student knowledge from observations (and why you should care)," in *Proceedings of the 13th International Conference of Artificial Intelligence in Education*, 2007, pp. 21–30.

[4] A. J. Martin and G. Lazendic, "Computer-adaptive testing: Implications for students' achievement, motivation, engagement, and subjective test experience." *Journal of Educational Psychology*, vol. 110, no. 1, p. 27, 2018.

[5] S. Yang, H. Wei, H. Ma, Y. Tian, X. Zhang, Y. Cao, and Y. Jin, "Cognitive diagnosis-based personalized exercise group assembly via a multi-objective evolutionary algorithm," *IEEE Transactions on Emerging Topics in Computational Intelligence*, vol. 7, no. 3, pp. 829–844, 2023.

[6] H. Ma, M. Li, L. Wu, H. Zhang, Y. Cao, X. Zhang, and X. Zhao, "Knowledge-sensed cognitive diagnosis for intelligent education platforms," in *Proceedings of the 31st ACM International Conference on Information & Knowledge Management*, 2022, pp. 1451–1460.

[7] Y. Zhou, Q. Liu, J. Wu, F. Wang, Z. Huang, W. Tong, H. Xiong, E. Chen, and J. Ma, "Modeling context-aware features for cognitive diagnosis in student learning," in *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*, 2021, pp. 2420–2428.

[8] X. Wang, C. Huang, J. Cai, and L. Chen, "Using knowledge concept aggregation towards accurate cognitive diagnosis," in *Proceedings of the 30th ACM International Conference on Information & Knowledge Management*, 2021, pp. 2010–2019.

[9] H. Ma, J. Zhu, S. Yang, Q. Liu, H. Zhang, X. Zhang, Y. Cao, and X. Zhao, "A prerequisite attention model for knowledge proficiency diagnosis of students," in *Proceedings of the 31st ACM International Conference on Information & Knowledge Management*, 2022, pp. 4304–4308.

[10] S. E. Embretson and S. P. Reise, *Item response theory*. Psychology Press, 2013.

[11] D. L. Torre and J., "Dina model and parameter estimation: A didactic," *Journal of Educational and Behavioral Statistics*, vol. 34, no. 1, pp. 115–130, 2009.

[12] F. Wang, Q. Liu, E. Chen, Z. Huang, Y. Chen, Y. Yin, Z. Huang, and S. Wang, "Neural cognitive diagnosis for intelligent education systems," in *Proceedings of the 2020 AAAI Conference on Artificial Intelligence*, vol. 34, no. 04, 2020, pp. 6153–6161.

[13] W. Gao, Q. Liu, Z. Huang, Y. Yin, H. Bi, M.-C. Wang, J. Ma, S. Wang, and Y. Su, "Rcd: Relation map driven cognitive diagnosis for intelligent education systems," in *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2021, pp. 501–510.

[14] G. Abdelrahman, Q. Wang, and B. Nunes, "Knowledge tracing: A survey," *ACM Computing Surveys*, vol. 55, no. 11, pp. 1–37, 2023.

[15] B. Zoph, V. Vasudevan, J. Shlens, and Q. V. Le, "Learning transferable architectures for scalable image recognition," in *Proceedings of the 2018 IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 8697–8710.

[16] H. Liu, K. Simonyan, and Y. Yang, "Darts: Differentiable architecture search," *arXiv preprint arXiv:1806.09055*, 2018.

[17] Y. Wang, Y. Yang, Y. Chen, J. Bai, C. Zhang, G. Su, X. Kou, Y. Tong, M. Yang, and L. Zhou, "Textnas: A neural architecture search space tailored for text representation," in *Proceedings of the 2020 AAAI Conference on Artificial Intelligence*, vol. 34, no. 05, 2020, pp. 9242–9249.

[18] A. Baruwa, M. Abisiga, I. Gbadegesin, and A. Fakunle, "Leveraging end-to-end speech recognition with neural architecture search," *arXiv preprint arXiv:1912.05946*, 2019.

[19] J. Chu, X. Yu, S. Yang, J. Qiu, and Q. Wang, "Architecture entropy sampling-based evolutionary neural architecture search and its application in osteoporosis diagnosis," *Complex & Intelligent Systems*, vol. 9, no. 1, pp. 213–231, 2023.

[20] M. D. Reckase, "Multidimensional item response theory models," in *Multidimensional item response theory*. Springer, 2009, pp. 79–112.

[21] Y. Koren, R. Bell, and C. Volinsky, "Matrix factorization techniques for recommender systems," *Computer*, vol. 42, no. 8, pp. 30–37, 2009.

[22] K. Xu, L. Han, Y. Tian, S. Yang, and X. Zhang, "Eq-net: Elastic quantization neural networks," *arXiv preprint arXiv:2308.07650*, 2023.

[23] T. Elsken, J. H. Metzen, and F. Hutter, "Neural architecture search: A survey," *The Journal of Machine Learning Research*, vol. 20, no. 1, pp. 1997–2017, 2019.

[24] B. Zoph and Q. V. Le, "Neural architecture search with reinforcement learning," *arXiv preprint arXiv:1611.01578*, 2016.

[25] J. Yu, P. Jin, H. Liu, G. Bender, P.-J. Kindermans, M. Tan, T. Huang, X. Song, R. Pang, and Q. Le, "Bignas: Scaling up neural architecture search with big single-stage models," in *Proceedings of the 2020 European Conference on Computer Vision*. Springer, 2020, pp. 702–717.

[26] B. M. Oloulade, J. Gao, J. Chen, T. Lyu, and R. Al-Sabri, "Graph neural architecture search: A survey," *Tsinghua Science and Technology*, vol. 27, no. 4, pp. 692–708, 2021.

[27] X. Su, S. You, J. Xie, M. Zheng, F. Wang, C. Qian, C. Zhang, X. Wang, and C. Xu, "Vision transformer architecture search," *arXiv preprint arXiv:2106.13700*, 2021.

[28] J. Xu, X. Tan, R. Luo, K. Song, J. Li, T. Qin, and T.-Y. Liu, "Nas-bert: Task-agnostic and adaptive-size bert compression with neural architecture search," in *Procedding of the 2021 ACM Conference on Knowledge Discovery and Data Mining*, 2021.

[29] X. Shi, P. Zhou, W. Chen, and L. Xie, "Darts-conformer: Towards efficient gradient-based neural architecture search for end-to-end asr," *arXiv preprint arXiv:2104.02868*, 2021.

[30] S. Yang, H. Ma, C. Zhen, Y. Tian, L. Zhang, Y. Jin, and X. Zhang, "Designing novel cognitive diagnosis models via evolutionary multi-objective neural architecture search," *arXiv preprint arXiv:2307.04429*, 2023.

[31] S. Yang, Y. Tian, X. Xiang, S. Peng, and X. Zhang, "Accelerating evolutionary neural architecture search via multifidelity evaluation," *IEEE Transactions on Cognitive and Developmental Systems*, vol. 14, no. 4, pp. 1778–1792, 2022.

[32] Y. Tian, S. Peng, S. Yang, X. Zhang, K. C. Tan, and Y. Jin, "Action command encoding for surrogate-assisted neural architecture search," *IEEE transactions on cognitive and developmental systems*, vol. 14, no. 3, pp. 1129–1142, 2021.

[33] D. E. Goldberg and K. Deb, "A comparative analysis of selection schemes used in genetic algorithms," in *Foundations of genetic algorithms*. Elsevier, 1991, vol. 1, pp. 69–93.

[34] K. Deb, A. Pratap, S. Agarwal, and T. Meyarivan, "A fast and elitist multiobjective genetic algorithm: Nsga-ii," *IEEE Transactions on Evolutionary Computation*, vol. 6, no. 2, pp. 182–197, 2002.

[35] S. Yang, Y. Tian, C. He, X. Zhang, K. C. Tan, and Y. Jin, "A gradient-guided evolutionary approach to training deep neural networks," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 33, no. 9, pp. 4861–4875, 2021.

[36] L. Zhang, H. Yang, S. Yang, and X. Zhang, "A macro-micro population-based co-evolutionary multi-objective algorithm for community detection in complex networks [research frontier]," *IEEE Computational Intelligence Magazine*, vol. 18, no. 3, pp. 69–86, 2023.

[37] A. P. Bradley, "The use of the area under the roc curve in the evaluation of machine learning algorithms," *Pattern Recognition*, vol. 30, no. 7, pp. 1145–1159, 1997.

[38] Y. Tian, S. Yang, and X. Zhang, "An evolutionary multiobjective optimization based fuzzy method for overlapping community detection," *IEEE Transactions on Fuzzy Systems*, vol. 28, no. 11, pp. 2841–2855, 2019.

Authorized licensed use limited to: Anhui University. Downloaded on October 26,2026 at 14:40:42 UTC from IEEE Xplore. Restrictions apply.

[39] Y. Bi, B. Xue, and M. Zhang, "Genetic programming with image-related operators and a flexible program structure for feature learning in image classification," *IEEE Transactions on Evolutionary Computation*, vol. 25, no. 1, pp. 87–101, 2020.
[40] H.-S. Chang, H.-J. Hsu, and K.-T. Chen, "Modeling exercise relationships in e-learning: A unified approach." in *Proceedings of the 2015 International Educational Data Mining Society*, 2015, pp. 532–535.
[41] AICFE, "Slp dataset," https://aic-fe.bnu.edu.cn/en/data/index.html, 2018.
[42] M. Feng, N. Heffernan, and K. Koedinger, "Addressing the assessment challenge with an online system that tutors as it assesses," *User Modeling and User-adapted Interaction*, vol. 19, no. 3, pp. 243–266, 2009.
[43] M. O. Z. San Pedro, R. S. d Baker, S. M. Gowda, and N. T. Heffernan, "Towards an understanding of affect and knowledge from student interaction with an intelligent tutoring system," in *Proceedings of the 2013 International Conference on Artificial Intelligence in Education*. Springer, 2013, pp. 41–50.
[44] D. P. . Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014.
[45] Y. . B. P . Vincent, H. Larochelle and P. .-A. Manzagol, "Extracting and composing robust features with denoising autoencoders," in *Proceedings of the 25th International Conference on Machine Learning*, 2008, p. 1096–1103.
[46] L. While, P. Hingston, L. Barone, and S. Huband, "A faster algorithm for calculating hypervolume," *IEEE Transactions on Evolutionary Computation*, vol. 10, no. 1, pp. 29–38, 2006.
[47] Y. Tian, J. Pan, S. Yang, X. Zhang, S. He, and Y. Jin, "Imperceptible and sparse adversarial attacks via a dual-population-based constrained evolutionary algorithm," *IEEE transactions on artificial intelligence*, vol. 4, no. 2, pp. 268–281, 2022.

Authorized licensed use limited to: Anhui University. Downloaded on October 26,2026 at 14:40:42 UTC from IEEE Xplore. Restrictions apply.